

BRIAN QIAN

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🐙 github.com/Azlothe

Education

University of Texas at Austin

Master of Science, Computer Science

Expected June 2027

GPA: 4.00/4.00

San Jose State University

Bachelor of Science, Computer Science

Graduated June 2024

GPA: 4.00/4.00

Experience

Company A

Start Date — Present

Software Engineer

Start Date – Present

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Software Engineer Intern

Start Date – End Date

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Company B

Start Date — End Date

Software Engineer Intern

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Projects

Tux Vision Language Model | *Python, PyTorch, Computer Vision, LoRA, Multimodal AI*

April 2026 — May 2026

- Fine-tuned SmolVLM-256M via **LoRA adapters**, achieving ~70% accuracy for visual question answering tasks on a Tux racing image dataset including object identification & recognition, spatial reasoning, and relative positioning.
- Built a **CLIP-style contrastive model** using the fine-tuned VLM's multimodal encoders to align image-text embeddings into a shared space, achieving a 21.4% accuracy improvement over the fine-tuned VLM model.

AWS Cloudathon | *ALB, ECS, DynamoDB, OpenSearch, S3, CloudFront, Lambda, VPC, CloudFormation*

April 2025

- Awarded 3rd place out of 100+ teams by designing and deploying an AWS stack with horizontal scaling, load balancing, zero-downtime deployment, deployment version control for snapshots/rollbacks, and OLAP system for analytics & BI.
- Architected and modernized an AWS architecture for a medium-sized business with \$10+ million portfolio, addressing key business concerns of production outages during traffic spikes, slow user experience, cost, and deployment failures.

Astrophysical N-Body Simulation | *C++, MPI, Distributed Systems, High-Performance Computing*

December 2025

- Built a distributed N-body simulator with **MPI**, partitioning 100K+ bodies across distributed nodes with replicated quadtrees and ring-based state exchange to eliminate centralized computation bottlenecks and serialization overhead.
- Reduced memory footprint by 80% (1.5 GB → 287 MB) after memory profiling with **valgrind**, identifying optimizations for the dynamic quadtrees through lazy quadrant allocation, incremental mass propagation, and precomputation.
- Investigated the communication-vs-computation tradeoff inherent in my design, analyzing empirical parallel performance to identify the inflection point where synchronization latency dominates computation throughput.

Hobbies & Interests: ...